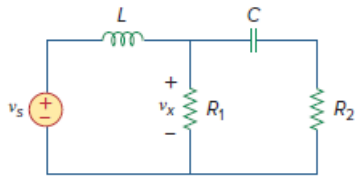


## Problem

Using Fig. 16.36, design a problem to help other students better understand circuit analysis using Laplace transforms.

**Figure 16.36:**



## Step-by-step solution

## Step 1 of 2

The voltage and current equations in time-domain will be in the form of differential equations because of the presence of inductor and capacitor, which are difficult to solve; while in the  $s$ -domain (Laplace domain), the equations will be in linear form, which are relatively easy to solve.

Transform the impedances associated with the elements into  $s$ -domain (Laplace domain).

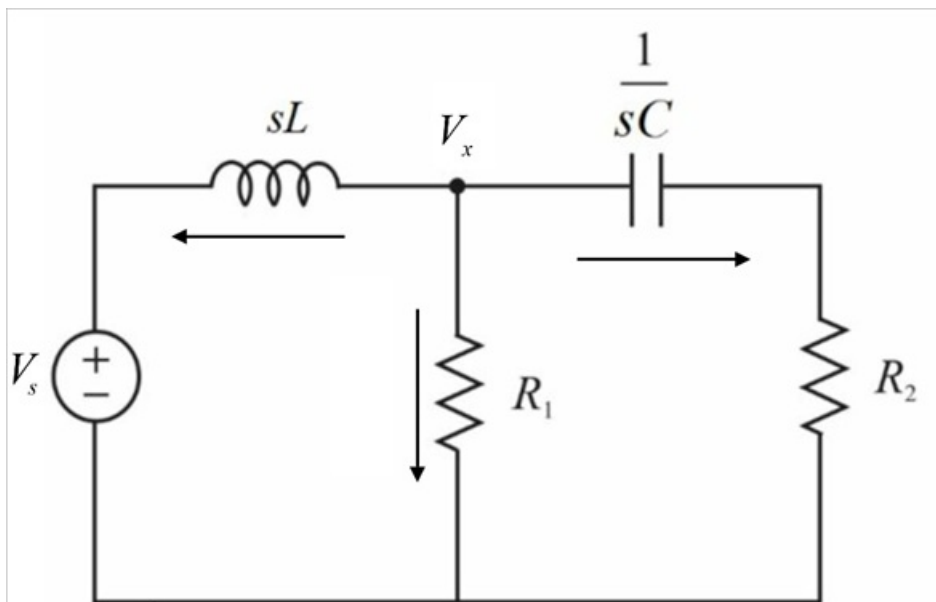
$$Z_L = sL$$

$$Z_C = \frac{1}{sC}$$

The resistances  $R_1$  and  $R_2$  remain the same in both the time domain as well as  $s$ -domain.

The Laplace transform of voltage source  $v_s$  is  $V_s$ .

Redraw the circuit in  $s$ -domain as shown in Figure 1.



**Figure 1**

## Step 2 of 2

Apply Kirchhoff's current law at node  $V_x$ .

$$\frac{V_x - V_s}{sL} + \frac{V_x}{R_1} + \frac{V_x}{\left(\frac{1}{sC} + R_2\right)} = 0$$

$$V_x \left( \frac{1}{sL} + \frac{1}{R_1} + \frac{sC}{1 + sR_2C} \right) = \frac{V_s}{sL}$$

$$V_x \left( \frac{R_1(1 + sR_2C) + sL(1 + sR_2C) + sC(sR_1L)}{sLR_1(1 + sR_2C)} \right) = \frac{V_s}{sL}$$

$$V_x \left( \frac{R_1 + sR_1R_2C + sL + s^2LR_2C + s^2CR_1L}{R_1(1 + sR_2C)} \right) = V_s$$

$$V_x \left( \frac{s^2(R_1LC + R_2LC) + s(L + R_1R_2C) + R_1}{R_1(1 + sR_2C)} \right) = V_s$$

$$V_x = \frac{R_1(1 + sR_2C)V_s}{s^2(R_1LC + R_2LC) + s(L + R_1R_2C) + R_1}$$

The value of  $V_x$  comprises of a linear equation in numerator and quadratic in the denominator, the roots of which are easy to determine.

The time-domain equivalent of  $V_x$ , that is,  $v_x$  is obtained applying inverse Laplace transform to  $V_x$ , which is the exact opposite of the Laplace transform.